

IKBKE fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes IKBKE gene fusions from the msk_impact_50k_2026 study, covering 12 fusion events. 3/12 fusions (25.0%) were in-frame. 4/12 fusions (33.3%) retained the PF00069 domain. Domain retention was tested with Fisher's exact test ($p=0.0242424$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for IKBKE.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was NAV1, observed in 2 events.

1.2 Domain retention

PF00069 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/3 (100.0%) of in-frame fusions retained the domain, $p=0.0242424$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p -value of 0.019802.

1.3 Domain disruption

Domain-disruption analysis was skipped: IKBKE has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 11 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 359 aa (permutation-corrected $p=0.39604$, not statistically significant at $\alpha=0.05$). This is 115 aa from the nearest configured domain boundary at 244 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: IKBKE has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p -value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: IKBKE has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
RASAL2	1	0.0833333333333333	0.17032913781186615		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.31083183493280936	1
NAV1	2	0.1666666666666666	0.17032913781186615		0.7275541795665634		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28816487341135866	2
NFASC	1	0.0833333333333333	0.17032913781186615		0.8885448916408669		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28694859742728085	3
TCEA3	1	0.0833333333333333	0.17032913781186615		0.5263157894736843		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.20932807553431312	4
RASSF5	1	0.0833333333333333	0.17032913781186615		0.44272445820433437		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.19141564740516673	5
SRGAP2	1	0.0833333333333333	0.17032913781186615		0.3405572755417957		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16952267969176557	6
RBBP5	1	0.0833333333333333	0.17032913781186615		0.26006191950464397		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.15227367482666163	7
THBS3	1	0.0833333333333333	0.17032913781186615		0.12693498452012386		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12374647447283589	8
SMYD2	1	0.0833333333333333	0.17032913781186615				['domain_retention', 'recurrence']	0.12287688082357551	9
IL20	1	0.0833333333333333	0.17032913781186615		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09654612064709506	10
RABGAP1L	1	0.0833333333333333	0.17032913781186615		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09654612064709506	11

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
77	0	4	1	6	0.0	1.0	-0.0
120	0	4	2	5	0.0	0.4909090909090909	0.30899892533525647
206	0	4	3	4	0.0	0.23636363636363636	0.626419337187407
359	0	4	4	3	0.0	0.19393939393939394	0.7123339658940003
395	1	3	4	3	0.25	0.5454545454545454	0.26324143477458145
447	3	1	4	3	2.25	1.0	-0.0
539	3	1	5	2	1.2	1.0	-0.0
572	3	1	6	1	0.5	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

3	1
0	7

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	11	4	3	4	0.1325227963525836

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
IL20	1
NAV1	2
NFASC	1
RABGAP1L	1
RASAL2	1
RASSF5	1
RBBP5	1
SMYD2	1
SRGAP2	1
TCEA3	1
THBS3	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint_pos	breakpoint_pos	protein_pos	protein_pos	domain_ref	domain_ref	domain_ref	disrupted_domain	disrupted_domain	annotation	effect_on_fra
EVT-P-0027193-T03-IM6-14	P-0027193-T03-IM6		IKBKE-IL20	IL20	unknown	three_prime	206667107	21	682	True	lost	0.0	False		Protein kinase domain		IL20 (NM_018724) - IKBKE (NM_014002) Rearrangement : c.398:IL20_c.2046-146:IKBKE Protein Fusion: mid-exon {IL20:IKBKE} IKBKE NA (NM_014002) - NAV1 (NM_020443) rearrangement: c.1340+100:IKBKE_c.1226+27872:NAV1 Protein Fusion: in frame {IK-BKE:NAV1}	
EVT-P-0051760-T01-IM6-15	P-0051760-T01-IM6		IKBKE-NAV1	NAV1	in-frame	five_prime	206653556	12	447	True	retained	1.0	False	Protein kinase domain			IKBKE NA (NM_014002) - NAV1 (NM_020443) rearrangement: c.1340+100:IKBKE_c.1226+27872:NAV1 Protein Fusion: in frame {IK-BKE:NAV1}	
EVT-P-0051760-T02-IM6-16	P-0051760-T02-IM6		IKBKE-NAV1	NAV1	in-frame	five_prime	206653558	12	447	True	retained	1.0	False	Protein kinase domain			IKBKE NA (NM_014002) - NAV1 (NM_020443) rearrangement: c.1340+102:IKBKE_c.1226+27873:NAV1 Protein Fusion: in frame {IK-BKE:NAV1}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_ratio	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0053124-T01-IM6-17	P-0053124-T01-IM6		IKBKE-RABGAP1L	RABGAP1L	out-of-frame	five_prime	206666746	20	682	True	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - RABGAP1L (NM_014857) rearrangement: c.2045+35:IKBKE_c.1710+64980:RABGAP1L	NA (2)	
EVT-P-0052992-T01-IM6-18	P-0052992-T01-IM6		IKBKE-RBBP5	RBBP5	out-of-frame	five_prime	206648412	5	120	True	disrupted	0.472340425	316149		Protein kinase domain		IKBKE (NM_014002) - RBBP5 (NM_005057) rearrangement: c.358+75:IKBKE_c.218+2153:RBBP5inv	NA (2)	
EVT-P-0058226-T01-IM6-21	P-0058226-T01-IM6		IKBKE-THBS3	THBS3	unknown	five_prime	206648064	5	77	True	disrupted	0.289361702	12765955		Protein kinase domain		IKBKE (NM_014002) - THBS3 (NM_007112) rearrangement: c.229-144:IKBKE_c.2579:THBS3inv	NA (2)	
EVT-P-0001902-T01-IM3-23	P-0001902-T01-IM3		NFASC-IKBKE	NFASC	in-frame	five_prime	206652479	10	395	True	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - NFASC (NM_001005388) Duplication: c.1183+3_c.2782+2783dup	NA (5388)	
EVT-P-0019364-T01-IM6-24	P-0019364-T01-IM6		RASAL2-IKBKE	RASAL2	unknown	three_prime	206652369	10	359	False	lost	0.0	False	Protein kinase domain			RASAL2 (NM_170692) - IKBKE (NM_014002) Rearrangement: c.1628-181:RASAL2_c.1076:IKBKEdel	NA (14002)	
EVT-P-0067348-T01-IM7-25	P-0067348-T01-IM7		RASSF5-IKBKE	RASSF5	out-of-frame	three_prime	206661210	16	539	True	lost	0.0	False	Protein kinase domain			mid-exon {RASAL2:IKBKE} RASSF5 (NM_182663) - IKBKE (NM_014002) rearrangement: c.579+1935:RASSF5_c.1617-41:IKBKEdup	NA (63)	
																	out of frame {RASSF5:IKBKE}		

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3 Figures

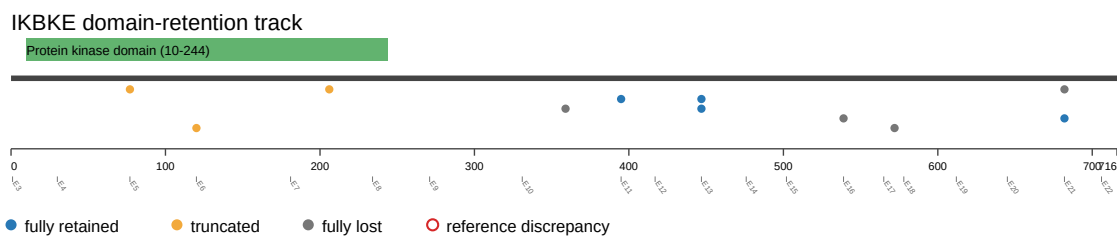


Figure 1: domain retention outliers

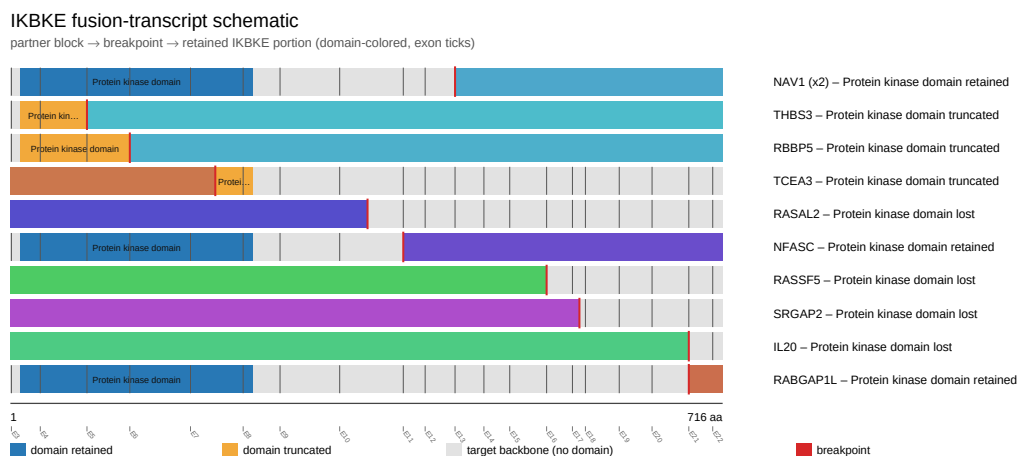


Figure 2: fusion schematic

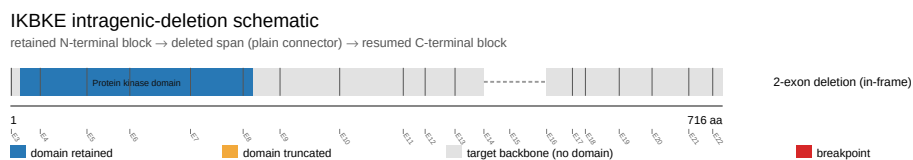


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison